

Priyanka Yadav

k.priyankatantla@gmail.com | [Google Scholar](#) | [LinkedIn](#) | [Webpage](#)

Contact Information

Department of Biotechnology
Indian Institute of Technology, Hyderabad
Telangana, 502284, India.

Mobile: +91-7568991775
Email: bo22resch11009@iith.ac.in
k.priyankatantla@gmail.com

Citizenship

Indian

Personal

- Born on August 01, 2000.
- Known languages are English and Hindi.

Area of Research

- Multiscale molecular modelling and simulation of Bio-nano systems

Systems of Interest

- Protein-drug and protein-protein interactions, ion channels (membrane proteins), nucleic acids and their interaction with drug molecules, topological DNA nanostructures, and solid-state nanopores.

Current Position

Senior Research Fellow at Department of Biotechnology, Indian Institute of Technology (IIT), Hyderabad, India, since July 2022.

- Advisor: [Dr. Himanshu Joshi](#)
- Thesis Title: Deciphering the structure and function of bio-nano systems using multiscale molecular simulation
- Area of study: Multiscale molecular simulations of biomolecular and bio-nano systems, focusing on the structure, dynamics, mechanics, and interactions of nucleic acids, membrane proteins, and nanopore systems.

Education

- M.Sc. (Life Science), 2021
[National Institute of Technology, Rourkela, India](#)
- B.Sc. (Botany Honours), 2019
[Maharani College, University of Rajasthan, Jaipur, Rajasthan, India](#)

Academic Honors

- Senior Research Fellowship awarded by CSIR in July, 2024.
- Qualified CSIR JRF, June 2021 (AIR 113)
- Qualified IIT-JAM in Biological Sciences, March 2019 (AIR 284)
- University Merit Position 5th in B.Sc. Honours Part 2nd

Software and Programming

- **Molecular Modelling Programs and methods:** GROMACS, NAMD, AMBER, Gaussian, FEP (alchemical free energy perturbation), Enhanced sampling simulations (SMD, Umbrella sampling)
- **Structure generation:** AlphaFold, CHARMM-GUI
- **Visualization Tools:** VMD, PyMol, Avogadro
- **Programming & OS:** Python, Tcl, NAB, Linux, Windows, Mac

Machine Learning Skills

- **Machine Learning Tools:** PyTorch, TensorFlow, Scikit-learn
- **Neural Network model:** PyTorch **multi-output model** for Combined Classification and Regression, Multi-Layer Perceptron (MLP)
- **Supervised machine learning models:** Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), Support Vector Machines /Regression (SVM/SVR), XG-Boost

Computational Chemistry & SBDD

- Structure-Based Drug Design (SBDD) of antiviral compounds against HIV, SARS-CoV-2.
- Familiar with DFT, molecular docking (AutoDock Vina), cheminformatics (RDKit, SMILES and Group graph)

Internships

IIIT Hyderabad, India (Visiting PhD scholar) – March 2023- Present

- Improving the CGenFF for small drug like molecules by using machine learning algorithms **under in supervision of Prof U. Deva Priyakumar**

Nihon Nohyaku Co., Ltd. (NNC), Osaka, Japan. December 08-20, 2025

- Structure-based Drug Designing Strategies (SBDD) in the Development of Novel Drugs.

NIT Rourkela, Odisha, India. July 2020 to December 2022

- Docking study of Tenofovir, Emtricitabine, Zidovudine, and Lamivudine against RNA-dependent RNA polymerase of SARS-Cov-2 and Reverse transcriptase of HIV-2, under the supervision of Dr. Suman Jha.

Publications

- Yadav P, Malik I, and Joshi H. “[Folding pathways and force-induced unfolding of neurodegeneration-associated GGGGCC microsatellite repeat RNA revealed by molecular simulations](#)”. International Journal of Biological Macromolecules, 372 (2026), 153101.
- Gupta S, Yadav P, Joshi H, “[SketchDNA: A GUI-Enabled Toolkit for Multiscale Modeling of Topological DNA Structures](#)”, BioRxiv. 2026:2026-07.

- Dahiya P, Yadav P, Singh A, Aggarwal S, Joshi H, Munde M. [“Protamine and Zinc Binding-Induced Conformational Effects on Insulin Analogs: An Integrated Experimental and Computational Study”](#). Physical Chemistry Chemical Physics, 2025, 27 (36), 19149–19162.
- D. Majhi, P. Yadav, Sahu PK, J Venkatesan, C Subrahmanyam. [“Chitosan-Based Nanoparticles in the Treatment of Alzheimer’s Disease”](#). Nanomedicine in the Treatment and Management of Alzheimer's Disease, pp. 137–151. CRC Press, 2025.
- Prajapati A. Srivastava A, Sankaranarayanan A, Pebam M, Buddhiraju HS, Patra P, Yadav P, Joshi H, Rengan AK. [“High-Affinity DNA Nanomatrix: A Platform Technology for Synergistic Drug Delivery and Photothermal Therapy”](#). ACS Macro Letters, 2023 12 (2),255-262.

Manuscript
Under Process

- D. Srivastava, P. Yadav, V. Nayak, V. Patil, A. Sharma, P. Mishra, H. Joshi, and A. Mishra, **“Integrated in-silico and experimental analysis identifies bromocriptine and lifitegrast as OTUB1-targeting compounds in B-cell acute lymphoblastic leukemia”**. *Manuscript under Review in Medical Oncology*.
- Yadav P, Zehra M, and Joshi H. **“Characterizing the conformational dynamics and ions permeation through Human NaV1.6 Sodium Channel”**. *Manuscript in preparation*.
- Yadav P, Malik I and Joshi H. **“Computational Investigation of Small-Molecule Binding to the C9orf72 GGGGCC RNA G-Quadruplex”**. *Manuscript in preparation*.
- Yadav P and Joshi H. **“Elucidate the molecular mechanism of electric field-driven ferritin capture and translocation through silicon nitride nanopores”**. *Manuscript in preparation*.
- Yadav P, Tadanki AS, Modee R, H Joshi and Priyakumar D. **“Machine Learning-Based Refinement of CGenFF Parameters for Enhanced Accuracy in Drug-Like Molecular Simulations”**. *Manuscript in preparation*.
- Chakraborty S, Kannihalli A, Yadav P, Das S, K B Yahwan, Aggarwal S, Joshi H, and Ray S. **“Circadian Regulation and Beta-Blocker Action: A Bidirectional Interplay Revealed by Metabolomics and Computational Modelling”**. *Manuscript in preparation*.

Scientific
Talks

- “Towards the fast and accurate force field parameters for small molecules using machine learning methods”. ML4Science conference, Trivandrum, Kerala, India. February, 2024.
- “Folding and unfolding of repeat RNA (G4C2) using MD simulations”. Horizons in Structural and Computational Biology, 2025, conference at IIT Hyderabad, India. March, 01, 2025.

Poster in
Various
Conference

- “Characterizing the folding and unfolding of G4C2 RNA repeats using a “computational microscope”. Institute of Protein Research, University of Osaka, Japan. December, 11, 2025.
- “BPS 2026”: Biophysical Society, an international conference in San Francisco, USA, from Feb 21-25, 2026.
- “IBS 2025: 47th international conference” at IIT Madras, India, from March 6-9, 2025.
- CSIR-Integrated skill program in “Emerging Topics in Molecular Dynamics Simulations” at CSIR National Chemical Laboratory, Pune, India, from January 20-24, 2025.
- “IBS 2024: 46th Indian Biophysical Society (IBS) international conference” at University of Hyderabad and TIFR Hyderabad, India, from March 15-17, 2024.
- “MD@60: International conference on MD simulations” at JNC SAR, Bangalore, India, from February 25-29, 2024.
- “ML4Science conference 2024”, Thiruvananthapuram, India, February 01-04, 2024

References

- **Dr. Himanshu Joshi**
Department of Biotechnology,
Indian Institute of Technology, Hyderabad
502284, India.
Email: hjoshi@bt.iith.ac.in
Website: [Molecular Simulation Lab](#)
- **Dr. Indranil Malik**
Department of Biotechnology,
Indian Institute of Technology, Hyderabad
502284, India.
Email: indranil@bt.iith.ac.in
Website: [The Malik Lab](#)